

DNA Replication & Repair

Current 35-page faculty deck primary - official Week 4 objectives integrated - replication pharmacology linked to lesion-specific repair

5 official objectives

Polymerases

Topoisomerases

Telomerase

DNA damage

Repair diseases

SOURCE PRIORITY: the uploaded current faculty deck controls terminology, examples, and emphasis. The official Week 4 learning-objective sheet defines what must be mastered. This companion reorganizes the same material for retrieval and application without replacing the source deck.

Official Week 4 Objectives

1. Describe how DNA-replication processes can be targeted by anticancer and antimicrobial drugs.
2. Explain RNA-dependent DNA replication in telomere synthesis and relate telomere dynamics to aging.
3. Name major DNA mutation types and describe endogenous and exogenous agents that cause them.
4. Summarize direct repair, base-excision repair, nucleotide-excision repair, mismatch repair, and recombination-mediated repair; connect repair defects to disease.
5. Explain why DNA-repair genes function as tumor-suppressor genes and why defective repair promotes cancer.

ASSIGNED SCHOLARRX

DNA Replication

DNA Mutation & Repair

Trinucleotide Repeat Disorders

Slides 3-7

Replication framework and polymerase logic

LO 1

- Replication occurs in S phase and is semiconservative: each daughter duplex contains one parental strand and one newly synthesized strand.
- Prokaryotes use a single principal origin; eukaryotes use multiple origins. Replication proceeds bidirectionally from each origin.
- Helicase separates strands using ATP. DNA polymerase extends only from a pre-existing 3'-OH and synthesizes 5' to 3'.
- Leading strand is continuous; lagging strand is discontinuous and contains Okazaki fragments that must be joined by ligase.

Polymerase / factor	High-yield job
Prokaryotic Pol III	Main high-processivity genome replicase; sliding-clamp associated
Prokaryotic Pol I	Primer removal + gap filling/repair; includes 5' to 3' exonuclease activity
Euk Pol alpha	Primase-associated initiation
Euk Pol beta	DNA repair
Euk Pol gamma	Mitochondrial DNA replication
Euk Pol delta	Lagging-strand synthesis
Euk Pol epsilon	Leading-strand synthesis
PCNA	Eukaryotic sliding clamp / processivity factor; high in proliferating cells

Proofreading is 3' to 5' exonuclease activity. The wrong terminal nucleotide is removed so synthesis can continue. Eukaryotic RNA primers are removed by RNase H.

Chain termination

DNA polymerase adds the incoming nucleotide to the **3'-OH** of the growing strand. Nucleoside analogs lacking a usable 3'-OH can terminate elongation.

Topoisomerase trapping

Replication creates positive supercoils ahead of the fork. Topoisomerases transiently break DNA, allow strand rotation/passage, and reseal it.

Target	Drug examples from deck	Consequence
Bacterial gyrase / topo II	Ciprofloxacin, novobiocin	Replication/transcription block; DNA double-strand breaks accumulate
Human topo II	Doxorubicin, etoposide	Stabilize cleavage complex after DNA break; DSBs
Human topo I	Camptothecin	Single-strand breaks

Topo I versus II: type I makes a single-strand nick and is ATP independent; type II breaks both strands, relaxes two supercoils at a time, and uses ATP.

Linear chromosome end

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Terminal RNA primer removed

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No upstream 3'-OH to fill final gap

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Telomere shortens without telomerase

Telomerase is a ribonucleoprotein reverse transcriptase: its RNA component provides the template, and its protein component catalyzes RNA-dependent DNA synthesis to extend chromosome ends.

Somatic cells

The lecture emphasizes low/absent telomerase in most somatic cells, progressive telomere shortening, and replicative senescence after a finite number of divisions.

Stem/cancer cells

Stem cells maintain telomerase. The deck notes telomerase activity in a large majority of human cancers, supporting continued proliferation and making telomerase an anticancer target.

Telomerase is RNA-dependent DNA polymerase. It is not an RNA polymerase and does not copy DNA into RNA.

Slides 12-20

Mutation types and sources of DNA damage

LO 3

Source	Damage / mutation pattern
Replication mismatch	Wrong base; microsatellites are hotspots
Trinucleotide-repeat hairpin/loop	Repeat expansion; anticipation
Spontaneous deamination	C to U can be recognized; 5-methyl-C to T is harder to detect and creates a mutation hotspot
Loss of base	Abasic site
Transposon insertion	Insertional disruption
Ionizing radiation	ROS, base loss, single- and double-strand breaks
UV radiation	Adjacent pyrimidine dimers, especially thymine dimers
Alkylating agent	Covalent alkyl groups on bases
Crosslinker / cisplatin	Intra/interstrand crosslinking and strand damage
Intercalator	Insertion/deletion mutations
Benzopyrene metabolite	Covalent guanine adduct

Ames test: a histidine-requiring Salmonella tester strain is exposed to a chemical. A mutagen raises the number of revertant colonies able to grow without supplied histidine.

Location matters: regulatory-region mutations can alter expression, coding-region mutations can alter protein sequence/function, and large DNA breaks can disrupt substantial genomic information.

Slides 21-32

DNA repair pathways: recognize the lesion, then choose the right repair logic

LO 4

General repair logic: identify the lesion, remove or reverse damaged material, restore correct sequence with DNA synthesis when needed, and ligate the backbone.

Pathway	Best lesion anchor	Mechanism / key proteins
Direct reversal	O6-alkylguanine	MGMT transfers alkyl group to itself and is then inactivated/degraded
Mismatch repair (MMR)	Replication mismatch; small insertion/deletion loop	MLH1, MSH2, MSH6, PMS2, EPCAM-associated pathway; scans for mismatches after replication
Base excision repair (BER)	Small base lesion, deamination, oxidation, SSB	DNA glycosylase removes base → AP endonuclease/ lyase remove abasic sugar-phosphate → polymerase + ligase restore strand
Nucleotide excision repair (NER)	Bulky helix distortion, thymine dimer, bulky adduct	Excises a short oligonucleotide containing lesion; polymerase fills; ligase seals
Homologous recombination (HR)	Double-strand break with sister-template available	Conservative, high-fidelity template-directed repair; late S/G2; BRCA1/2 promote HR
Non-homologous end joining (NHEJ)	Double-strand break without template	End binding/trimming + ligation; faster but error prone; important in G1

Translesion synthesis (TLS)	Replication-blocking lesion	Specialized bypass synthesis lets replication continue across damaged template
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BER = bad base. NER = bulky distortion. MMR = replication mismatch. HR/NHEJ = double-strand break.

Slides 24-32

Repair defects and disease associations

LO 4-5

Defect	Result	Disease anchor
MMR loss	Microsatellite instability	Lynch syndrome ; especially colorectal and endometrial cancer risk, with multiple other cancer risks
Global NER loss	Failure to repair UV-induced/bulky lesions throughout genome	Xeroderma pigmentosum ; extreme UV sensitivity, freckling/skin damage, markedly increased skin-cancer risk
Transcription-coupled NER loss	Failure to repair lesions blocking active transcription	Cockayne syndrome
BRCA1/BRCA2 pathway dysfunction	Reduced high-fidelity HR of DSBs	Genome instability / inherited cancer susceptibility

MMR disease logic: mismatch repair defects do not create one specific mutation. They create a *mutator phenotype*, especially microsatellite instability, so many genes across multiple tissues can acquire pathogenic mutations.

Slides 33-34

Why DNA-repair genes are tumor suppressors

LO 5

Cells experience enormous numbers of DNA modifications. Repair pathways normally prevent these lesions from becoming fixed mutations. When damage is substantial, checkpoint systems can pause the cell cycle; p53-centered pathways can promote senescence or apoptosis if damage cannot be safely repaired.

Repair-gene loss

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Mutation burden rises

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Genome instability

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Oncogenic/tumor-suppressor hits accumulate

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Cancer risk rises

Repair proteins are "guardians of the genome." Their normal function suppresses tumor formation; loss of function therefore behaves as tumor-suppressor loss.

Exam traps

Keep pathway and enzyme directionality straight

All LOs

Trap 1: DNA synthesis is 5' to 3'; proofreading is 3' to 5' exonuclease.

Trap 2: Topo I creates SSBs; topo II creates DSBs.

Trap 3: Telomerase uses RNA as a template to synthesize DNA.

Trap 4: BER removes a damaged base first; NER removes an oligonucleotide around a bulky lesion.

Trap 5: HR is high fidelity and template dependent; NHEJ is template independent and more error prone.

Trap 6: Lynch = MMR; XP/Cockayne = NER; BRCA1/2 = HR support.

30-SECOND FINISH

3'-OH required for extension; 3' to 5' proofreads. Topo inhibitors trap DNA breaks. Telomerase solves chromosome-end replication. UV makes pyrimidine dimers. MMR fixes replication errors, BER fixes small base lesions, NER fixes bulky distortions, HR/NHEJ fix DSBs. Repair-gene loss increases genome instability and cancer risk.